Advanced Machine Learning for Imaging and Vision
A.Y. 2025/2026

Lab 3: An Image Search Engine using CLIP

Lab duration: 6h

*Report due before **May 19th h23.59**. Late reports will not be graded.*
Report must be uploaded to "Portale della Didattica", "Elaborati" section as a group_n.zip file with report.pdf and code.ipynb as exported from Google Colab.
This lab is worth up to 2 points.

Download the starting notebook from Portale della Didattica.

In this lab you will build an image search engine, similar to Google Images, using the powerful multimodal CLIP model. A collection of images from the Flickr8k dataset is given to you. Your search engine will be able to search through this collection to:

- find matching images given a text query;
- find similar images given an image query.

You will be using the pretrained CLIP model provided by OpenAI (https://github.com/openai/CLIP.git). You will need to read the documentation of the Github repository to understand how to interact with the model. Some important notes:

- the CLIP code provides the image and text encoders, as well as an image preprocessor to prepare image to be fed to the image encoder, and a text tokenizer;
- CLIP text/image encoders return features as half-precision floating point values (float16). If you want to perform computations involving other precisions (e.g. float32), then you must cast all to have the same type (all float16 or all float32).

Creating a database of image features

Given a collection of images, we can use the CLIP image preprocessor and image encoder to encode each of them as a vector of features (e.g. 512 features), creating a database of representations of the known images. If a new image or a new text are semantically similar they will be encoded but the CLIP image/text encoders into feature vectors similar to the ones in the database.

Retrieving an image from a text query

Given a text query is the form of a string (e.g. “two dogs”), the text must be first tokenized. Then its encoding is produced by the CLIP text encoder as a vector of features. We then compute its cosine similarity with all the vectors in the database to find which images in our collection are most similar (higher cosine similarity of the encodings).

Retrieving an image from an image query

Given an image query we use the CLIP image preprocessor and image encoder to encode each of them as a vector of features. We then compute its cosine similarity with all the vectors in the database to find which images in our collection are most similar (higher cosine similarity of the encodings).

Your report must explain how you used CLIP to build the engine and show the following results:

- The 5 images best matching the query “*A boat*”
- The 5 images best matching *1786425974_c7c5ad6aa1.jpg*
- Estimate how big is your database in terms of memory occupation, considering the size of the image features extracted by CLIP and the number of images in the collection.